

Leveraging Social Media Analysis for Effective Water Management

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ABSTRACT

This article explores the potential of social media as a tool for water management. With the increasing challenges of climate change, droughts, and social inequities, the need for efficient water management is critical. However, social media is underutilized in this regard. The paper discusses the opportunities for social media to collect and disseminate information about water concerns and examines a data set of water utility tweets using AI and Machine Learning tools. The study concludes that social media is crucial for addressing global issues, and AI analytics can guide the effective use of social media for positive change.

CCS CONCEPTS

• **Information systems** → **Content analysis and feature selection**; • **Computing methodologies** → **Machine learning**; • **Applied computing** → **Document management and text processing**.

KEYWORDS

Water Management, Bert, Spherical k-means, Classification, Clustering

ACM Reference Format:

Khalid K. Osman, Florian Barbaro, and Andy Skumanich. 2023. Leveraging Social Media Analysis for Effective Water Management. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3584931.3607003>

1 INTRODUCTION

There are multiple aspects of using social media for water management which range from customer sentiment (such as direct feedback about water quality) to more regional or global aspects of *water anxiety*. The former can be part of a water utility's engagement with its customer base, and social media provides a useful (although underutilized) path of communication. However, the latter aspect of regional water concerns is of increasing importance given the developing global water supply issues. We focus on the latter.

*All three authors contributed equally to this research.

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CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA

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ACM ISBN 979-8-4007-0129-0/23/10...\$15.00

<https://doi.org/10.1145/3584931.3607003>

The water utilities need to expand the immediate outreach for the benefit of their customers [16]. In this paper, we explain how high-level data analysis can show regional water aspects which can be directed to assist the communities. The social media-derived data can be monitored and addressed to improve the overall functioning of water infrastructure management for the benefit of society in challenging times [12]. We suggest that the proper use of social media can assist in that endeavour.

A recent publication [8] focused on how water utilities use social media to reach their customers and determined that the water sector was behind other industries, including the energy sector, in using social media. This study analyzed the social media profiles of utilities across the United States and showed that only a tiny percentage of them were using social media, and an even smaller fraction reached their customer base. The range of engagement varied widely and highlighted the challenge of using social media. We propose that this shortcoming needs to be addressed because of the potential value of social media for not just monitoring customer satisfaction but for general water management using the community as a partner. This paper focuses on the more global aspect of social media as reflecting *water anxiety* or other social concerns from a more macro perspective. We find that by using a refined AI/ML approach along with interdisciplinary social science, we can extract regional expressions which can be important indicators of community engagement. Monitoring and influencing social awareness and attitudes can lead to improved overall water management strategies.

2 METHODS

This section describes the overall approach of the proposed application. The details of the implementation are presented next.

2.1 Data

2.1.1 Dataset collection. The data consisted of water utilities' tweets in the aftermath of the COVID-19 pandemic. Tweets along with engagement data (likes, retweets) were extracted directly from water utilities' Twitter pages utilizing a Twitter Developer Application Programming Interface (API) [14]. The water utilities examined in this study were selected based on whether their county met the US Center for Disease Control (CDC) hotspot classification [17] at least one time between March 2020 and October 2020 for three states, that is to say, California, Florida and Utah. It was critical that we captured data early in the pandemic so that we are analyzing crisis response immediately or shortly after the pandemic began. Water utilities within the counties were then identified by manually searching county websites. Over 25000 tweets were extracted from 40 water utilities from January 01, 2020 (World Health Organization

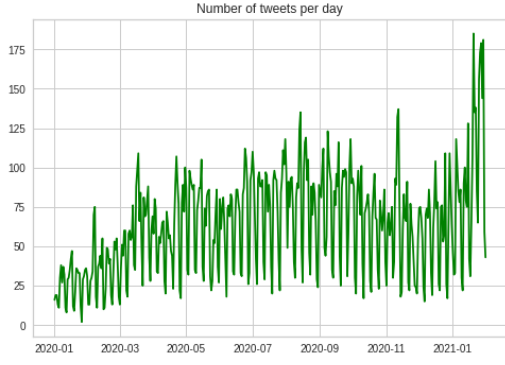


Figure 1: Number of Tweets per day for the three states.

declared COVID-19 a public health emergency on January 30, 2020 [5]) to January 30, 2021. Table 1 shows the distribution of tweets and utilities by state. The database is heterogeneous containing mostly Tweets from counties in Florida and shows the difference in the propagation of Covid in each state.

Each Twitter account for the utilities was manually verified by visiting each profile as many counties had multiple utilities. Utilities without a Twitter presence were excluded from the study. Additionally, some of the tweets regarding water utilities were posted by town, city, or county-level Twitter accounts rather than the utilities themselves. This was particularly true for smaller counties where everything was managed by one entity. These tweets remained in the study and were treated with the same veracity as those from utilities.

Table 1: Distribution of tweets and utilities by state

State	Nb. of tweets	Nb. of utilities
California	5521	13
Florida	17957	21
Utah	1657	6
Total	25135	40

Figure 1 shows the Number of tweets per day. We can see a clear increase in the number of tweets over time.

2.1.2 Data preprocessing. We used two distinct strategies for text processing depending on the algorithm used. For BERT, we kept the text as it was. However, for other analyses, we lowercase the text to apply a traditional pre-processing by removing stop-words and digits, and lemmatisation [2]. Then we kept the N-grams from $N = 1$ to $N = 4$ and removed the rare ones (appearing in less than 1% of documents) to obtain 8242 N-grams. Finally, we used the term frequency-inverse document frequency (TF-IDF) to weigh the N-grams.

2.2 Models

2.2.1 Spherical k-means (Sk-means). Many models are poorly suited to high-dimensional data, including those derived from the vector representation of text [3]. When the data is directional [15], i.e.

when it is their correlation rather than their Euclidean distance that matters, Gaussian-type models are even less suitable. For such data, it is natural to carry out a normalisation that places them on the unity sphere. Then one can use clustering techniques that address specifically the fact the data are spherical, such as spherical k-means [7].

The spherical k-means algorithm (Sk-means), originally proposed in [7], is a simple adaptation of the k-means algorithm to the cosine dissimilarity. Let us consider a collection of N observations $X = (\mathbf{x}_i)_{1 \leq i \leq N}$ on the hypersphere $\mathbb{S}^{d-1}$. Given a number of clusters K , Sk-means tries to find a set of K prototypes $(\boldsymbol{\mu}_k)_{1 \leq k \leq K}$ in $\mathbb{S}^{d-1}$ and a clustering/membership $Z = (z_i)_{1 \leq i \leq N}$, that assigns $\mathbf{x}_i$ to cluster $z_i \in \{1, \dots, K\}$ such that the coherence

$$Q((\boldsymbol{\mu}_k)_{1 \leq k \leq K}, (z_i)_{1 \leq i \leq N}) = \sum_{i=1}^N \boldsymbol{\mu}_{k_i}^T \mathbf{x}_i \quad (1)$$

is maximal (see e.g. [11]).

In the following experience, we used the implementation of [13], which proposes a fast initialization and enforces sparsity on the centroid vectors by using a data-driven threshold that is capable of dynamically adjusting its value depending on the clusters.

2.2.2 BERT. Bidirectional Encoder Representations from Transformers (BERT) is an embedding layer designed to train deep bidirectional representations from unlabeled texts by jointly conditioning on both left and right context in all layers. The BERT model consists of two steps: pre-training and fine-tuning [6]. During pre-training, the model is trained on an unlabeled large corpus. For fine-tuning, the model is initialized with the pre-trained parameters and all the parameters are fine-tuned using labelled data for specific tasks. BERT's model architecture is a multi-layer bidirectional Transformer encoder [6] based on the original implementation described in [18]. This kind of encoder is composed of a stack of $N = 6$ identical layers. Each of these layers has two sub-layers. The first one is a multi-head self-attention mechanism, and the second one is a simple position-wise fully connected feedforward network. It employs a residual connection [10] around both sub-layers, followed by a layer normalization [18]. In the following experience, we used the model pre-trained in [9] to tackle the disinformation about Covid based on bert-large-uncased.

3 RESULTS

3.1 Clustering using Sk-means

First of all, the main problem when using clustering models is selecting the number of clusters. To do so, we used the Elbow method and got $K=8$.

Then we applied the Spherical k-means to the water management corpus focusing on three states: Florida, California and Utah. Table 2 shows the distribution of reports by cluster obtained. The distribution of Tweets in each cluster is quite heterogeneous with two important clusters, 2 and 8, a minor cluster, 3, and the rest of the clusters which are quite homogeneous between them.

Thanks to the implementation of sk-kmeans from [13] which enforces the sparsity on the centroid vectors, we are able to retrieve the most important N-grams by cluster. The table 3 shows the top 5 N-grams for each cluster. It is interesting to note that these N-

Table 2: Distribution of tweets by cluster

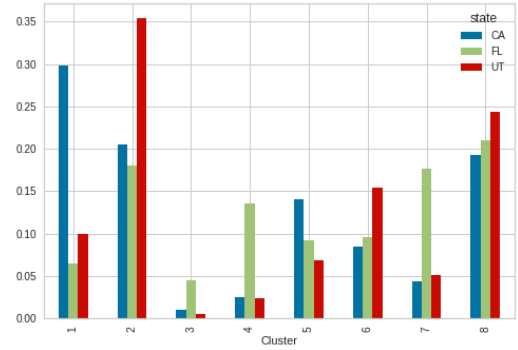
Clusters	Nb. of News	Clusters	Nb. of News
1	2964	5	2534
2	4965	6	2446
3	879	7	3500
4	2604	8	5243

Table 3: Top 5 words by cluster for Sk-means

Clusters	Top words
cluster 1	water supply, water news, water education, save water, emergency notifications
cluster 2	city commission, citystaugtbr, citystaug citystaugtbr, convene, city commission meeting
cluster 3	latest report, latest report visit, view latest report, view latest report visit, positivity rate
cluster 4	local covid case, update local covid case, local covid case currently, currently hospitalized death highlandscounty, death highlandscounty
cluster 5	power restored, restore power, experiencing power, experiencing power outage, restoration time
cluster 6	take census, makeitcount, census today, fill census, census online
cluster 7	board county commissioner, covid statistic, county covid statistic, seminole county covid, seminole county covid statistic
cluster 8	stop spread, help slow, help slow spread, wear mask, help slow spread covid

grams defining clusters express notions that are essential to the risks faced by Water Management utilities. Cluster 1 is related to water supply and related environmental issues. Cluster 2 pertains to city commissions or other forms of local government meetings. It appears that these entities are convening or discussing various issues related to their communities. Clusters 3, 4, 7 and 8 discussed Covid related efforts deployed by their utilities or counties to stop the spread of the illness and keep people up to date on the situation. Cluster 5 is related to power outages which could disrupt daily life activities and can cause problems with Water Management. People are asked to keep abreast of any news about the restoration efforts of the utilities concerned. The focus for Cluster 6 appears centred around census initiatives, suggesting citizens have been encouraged by their governments/local leaders to take part in this year's count; they're also being implored online using hashtags (#MakeItCount). As well, respondents might be directed towards specific websites where they can fill out details digitally instead of having face-face interactions during door-knocking campaigns traditionally employed by enumerators collecting data manually across neighbourhoods nationwide annually every decade since 1790.

Figure 2 shows the strength of a cluster for the 3 states. As we can observe, all clusters occur in each state, however with notable differences. California, as mentioned earlier, is more concerned with water supply as shown in Cluster 1 and therefore needs to communicate on the many issues that may arise with Cluster 5.

**Figure 2: Distribution of State by cluster.**

Florida, which is the strongest in clusters 3, 4 and 7, is focusing on monitoring the evolution of Covid and informing the population. Utah focuses on the responsibility of informing its citizens about decisions made by public authorities as well as inviting its citizens to participate in the state's public life, as evidenced by clusters 2 and 6. Finally, it should be noted that these three states are comparable in regards to the health procedures for Covid independent of different political orientations, as shown by Cluster 8.

We note that despite the mix of several states with different water management policies, essential information for each state is not diluted and enables localized aspects to be obtained. This is the type of information which could be used by governments and regional authorities to direct messaging or resources to engage with the appropriate community. Given the limitation of resources, correct targeting can be essential. The fact that no *water anxiety* is showing up in the social media for UT is in itself a point of note. In UT there was no reflection of *water anxiety* even though there has been a significant drought in the just prior 2018-2019 time frame along with other water supply issues. Importantly, this is a major developing issue in that experts warn that the Great Salt Lake is set to disappear in the next five years [1].

3.2 Binary classification using BERT for information credibility

A critical aspect of social media is the level of credibility [4] and this is a separate focus of our research. To do so, we first fine-tuned BERT on a labelled database and then applied it to our data.

We used BERT fine-tuned on the database from [9]. They performed manual annotation of tweets about vaccine misinformation which was controlled by medical experts. Consequently, a total of 15,073 tweets were labelled, 5,751 of which were misinformation and 9,322 general vaccine-related tweets. Moreover, we followed the same procedure. We first trained and validated 75% of the dataset and then evaluated the remaining 25% of the dataset. BERT was trained for three iterations with a 20% validation set taken from a subset of the training set. Finally, we obtained the same results as presented in [9] with an accuracy and an F1-score of 98% on the test set.

We applied it to our database and kept only Covid tweets related, that is to say, 4,344 tweets. Then, we evaluated these tweets

regarding misinformation. In the cases of the water utilities' social media, there was no detected misinformation. Instead, the content provided useful and relevant information. Note that this finding is important in that it demonstrates that some sources of Covid information can be socially augmenting. The connection with water utilities is that the water sources can be post-monitored for viruses and can provide important metrics regarding infections and local or regional levels (which manifest in the tested wastewater). In principle, this information could be used as a local health alert.

4 DISCUSSION

In this article, we observed that with even a limited set of social media data, key societal and water issues can be extracted with the appropriate AI/ML analysis along with interdisciplinary social science. The clustering shows significant and meaningful results. California has dominant clusters associated with *water anxiety*, whereas Florida does not. Instead, Florida is associated with *Covid anxiety* which is an adjacent social issue. In the case of Utah, there is no *water anxiety* and reduced *Covid anxiety*. Utah does not have the same level of drought as California, however, it is developing and would become a major issue. It is important to note the fact that the Utah data did not show *water anxiety*. This *null* result can instead be considered indicative of the need for improved social communication about developing issues. The local utilities can recognize the need to ramp up messaging about drought and conservation. They can monitor the analytic results to see the success of the messaging into the community. We find that the utilities' information was high quality and could provide local health alerts.

The data we analyze shows how critical the use of social media will be not just at the customer relations level (not developed in detail in this paper), but at a global level looking at the aspects of, for instance, *water anxiety*. This type of analysis can allow governments and policy organizations to better detect and target optimization efforts. In addition, the appropriate use of social media can include crowd-sourcing and citizen scientist inclusion which is an invaluable asset for managing critical social challenges such as water distribution. This would be a starting point for socially inclusive data which can be very important for optimal management of a diminishing or valuable critical human needs resource.

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